GROKKING THROUGH COMPRESSION: UNVEILING SUDDEN GENERALIZATION VIA MINIMAL DESCRIPTION LENGTH

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Paper under double-blind review

ABSTRACT

This paper investigates the relationship between Minimal Description Length (MDL) and the phenomenon of grokking in neural networks, offering an information-theoretic perspective on sudden generalization. Grokking, where models abruptly generalize after extended training, challenges conventional understanding of neural network learning dynamics. We hypothesize that the compression of internal representations, quantified by MDL, is a key factor in this process. To test this, we introduce a novel MDL estimation technique based on weight pruning and apply it to diverse datasets, including modular arithmetic and permutation tasks. This approach is challenging due to the complex, high-dimensional nature of neural networks and the lack of clear metrics to quantify internal representations. Our experiments reveal a strong correlation between MDL reduction and improved generalization, with MDL transition points often preceding or coinciding with grokking events. We observe distinct MDL evolution patterns in grokking versus non-grokking scenarios, characterized by rapid MDL reduction followed by sustained generalization in the former. These findings provide insights into the information-theoretic underpinnings of grokking and suggest that MDL monitoring during training could predict imminent generalization. Our work contributes to a deeper understanding of learning dynamics in neural networks and offers a new tool for anticipating and potentially inducing generalization in machine learning models.

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1 INTRODUCTION

The field of deep learning has witnessed remarkable progress in recent years, with neural networks achieving unprecedented performance across various domains Goodfellow et al. (2016). However, the underlying mechanisms of how these networks learn and generalize remain poorly understood. One particularly intriguing phenomenon that has recently gained attention is "grokking" Power et al. (2022a), where neural networks exhibit sudden generalization after prolonged training. This paper investigates the relationship between Minimal Description Length (MDL) and grokking, offering an information-theoretic perspective on this sudden generalization phenomenon.

Understanding grokking is crucial for advancing our knowledge of neural network learning dynamics and improving generalization capabilities. However, explaining grokking presents significant challenges:

- It contradicts the conventional understanding of gradual learning in neural networks.
- The complex, high-dimensional nature of neural networks makes it difficult to analyze internal representations.
- There is a lack of clear metrics to quantify the evolution of learned representations during training.

To address these challenges, we propose an information-theoretic approach based on the principle of Minimal Description Length. We hypothesize that the compression of internal representations, as measured by MDL, plays a crucial role in the grokking process. Our approach involves:

- Implementing a novel MDL estimation technique using weight pruning.
- Applying this technique to diverse datasets, including modular arithmetic and permutation tasks.
- Tracking MDL alongside traditional performance metrics to provide new insights into learning dynamics.

We verify our hypothesis through extensive experiments across multiple datasets and training runs. Our analysis reveals:

- A strong correlation between MDL reduction and improved generalization.
- Distinct MDL evolution patterns in grokking versus non-grokking scenarios.
- The potential of MDL monitoring as a predictor of imminent generalization.

The main contributions of this paper are:

- A novel MDL estimation technique for neural networks based on weight pruning.
- Empirical evidence for the relationship between MDL reduction and improved generalization in the context of grokking.
- Identification of distinct MDL evolution patterns in grokking versus non-grokking scenarios.
- Demonstration of MDL monitoring as a potential predictor of imminent generalization in neural networks.

Our work opens up several avenues for future research, including:

- Exploring the relationship between MDL and grokking in more complex architectures and tasks.
- Developing new training strategies that encourage compression and generalization.
- Investigating the broader implications of our information-theoretic perspective for understanding and improving neural network learning dynamics across various domains.

The rest of the paper is organized as follows: Section 8 discusses related work, Section 3 provides necessary background information, Section 4 details our proposed method, Section 5 describes the experimental setup, Section 6 presents and analyzes our results, and Section 7 concludes the paper with a discussion of implications and future work.

2 Related Work

The phenomenon of grokking, first introduced and extensively studied by Power et al. (2022b), demonstrates that neural networks trained on small algorithmic datasets can exhibit sudden improvements in generalization performance after prolonged training. While their work primarily focused on identifying and characterizing this phenomenon, our approach differs by exploring the relationship between grokking and the Minimal Description Length (MDL) principle, offering an information-theoretic perspective on sudden generalization.

Goodfellow et al. (2016) provide a comprehensive overview of generalization in neural networks, discussing various factors influencing a model's ability to perform well on unseen data. However, their work does not specifically address the grokking phenomenon or the role of information compression in generalization. Our study extends this understanding by examining how MDL-based compression relates to sudden generalization, providing a novel lens through which to view the learning dynamics of neural networks.

The Information Bottleneck theory, proposed by Bahdanau et al. (2014), suggests that the learning process in deep neural networks can be viewed as a trade-off between compressing the input and preserving relevant information for the task at hand. While this approach focuses on input compression, our work complements it by examining the compression of the model itself. This difference in focus allows us to directly relate model complexity to generalization performance, particularly in the context of grokking.

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Paszke et al. (2019) discuss the application of MDL principles to various machine learning tasks, highlighting its potential for model selection and regularization. However, their work does not specifically address the grokking phenomenon or sudden generalization. Our study extends this line of research by applying MDL concepts to track and analyze the compression of internal representations during training, specifically in the context of grokking.

Recent work by Radford et al. (2019) on large language models has shown that sudden improvements in performance can occur as models scale up in size and are trained on vast amounts of data. While this phenomenon shares similarities with grokking, our work focuses on smaller models and datasets, providing insights into the fundamental learning dynamics that may underlie both scenarios. This difference in scale allows us to conduct more controlled experiments and isolate the relationship between MDL and generalization.

Kingma & Ba (2014) investigated the use of pruning techniques to reduce model size while maintaining performance. Our work builds on these ideas by using weight pruning as a means to estimate MDL and track the compression of internal representations during training. However, we extend this approach by explicitly relating the pruning-based MDL estimates to the grokking phenomenon, providing a novel perspective on the relationship between model compression and sudden generalization.

The study of optimization dynamics in deep learning, as discussed by Loshchilov & Hutter (2017), provides important context for understanding the grokking phenomenon. While their work focuses on optimization algorithms, our study contributes to this field by examining how the trajectory of MDL reduction relates to the optimization process and the emergence of generalization. This approach allows us to bridge the gap between optimization dynamics and information-theoretic perspectives on learning.

Finally, while Vaswani et al. (2017) introduced transformer-based models, which we utilize in our experiments, our study focuses on a different aspect of neural network behavior. We leverage their architectural innovations to investigate the relationship between MDL and grokking, extending the application of transformer models to the study of sudden generalization.

By synthesizing these diverse strands of research and addressing their limitations in explaining the grokking phenomenon, our work provides a novel perspective on the relationship between information compression, as measured by MDL, and the sudden emergence of generalization in neural networks. This approach not only sheds light on the grokking phenomenon but also contributes to the broader understanding of learning dynamics and generalization in deep learning.

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3 BACKGROUND

Deep learning has revolutionized machine learning, achieving unprecedented performance across various domains Goodfellow et al. (2016). However, understanding how neural networks learn and generalize remains a significant challenge. Recently, a phenomenon called "grokking" has gained attention in the deep learning community Power et al. (2022a). Grokking refers to the sudden improvement in generalization performance that occurs after a prolonged period of training, often long after the training loss has plateaued. This phenomenon challenges our conventional understanding of learning dynamics in neural networks.

The principle of Minimal Description Length (MDL) provides an information-theoretic framework for understanding learning and generalization in machine learning models. Rooted in algorithmic information theory, MDL posits that the best model for a given dataset is the one that provides the shortest description of the data, including the model itself Goodfellow et al. (2016). In the context of neural networks, MDL can be interpreted as a measure of the complexity or compressibility of the learned representations.

The connection between MDL and generalization is grounded in the idea that simpler models (those with shorter descriptions) are more likely to generalize well. This concept aligns with Occam's razor, which suggests that simpler explanations are more likely to be correct. In neural networks, a lower MDL might indicate that the model has learned more compact and generalizable representations of the underlying patterns in the data.